

CIRCULAR COLOURINGS OF ABELIAN CAYLEY GRAPHS BELOW FOUR COME FROM CHARACTERS

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ABSTRACT. We show that a Cayley graph of an abelian group has a homomorphism to the circular clique $K_{p/q}$, for $p/q < 4$, if and only if some character maps every generator into the closed arc $[q/p, 1 - q/p]$ of $\mathbb{R}/\mathbb{Z}$. So its circular chromatic number, when it is less than 4, equals $1/\kappa(S)$, where $\kappa(S)$ is the largest r such that some character keeps every generator at distance at least r from 0; the bound 4 is sharp. In particular, the graph is 3-colourable if and only if some character maps every generator into $[1/3, 2/3]$, and it maps to the odd cycle C_{2k+1} if and only if some character maps the generators into $[k/(2k+1), (k+1)/(2k+1)]$. The proof averages the winding of circular colourings, known from work on their reconfiguration, over the group with an invariant mean. Consequences: (i) every set of Bohr recurrence is a set of 3-chromatic recurrence, the case of three colours of Katznelson’s question, and its Cayley graph has circular chromatic number at least 4; for every 3-colouring $\mathbb{N} = A_1 \cup A_2 \cup A_3$ the union $\bigcup_i (A_i - A_i)$ contains $\{n : \|n\alpha\| < 1/3\}$ for some α , which answers a question of Glasscock, Koutsogiannis and Richter; (ii) $\chi_c(G(\mathbb{Z}, D)) = 1/\kappa(D)$ for every distance graph with three distances, which answers a question in Liu’s survey on distance graphs; (iii) a Cayley graph of a finitely generated abelian group on a finite connection set with $\chi_c < 4$ has a periodic optimal circular colouring, and χ_c is computable below 4, while for many colours, by the theorem of Greenfeld and Tao on tilings, some Cayley graphs of $\mathbb{Z}^d$ have no periodic optimal colouring; (iv) a uniform proof that no Cayley graph of an abelian group of exponent 2 or 4 has circular chromatic number strictly between 2 and 4, which contains Payan’s theorem on cube-like graphs; (v) for a number field F , the plane F^2 is 3-colourable if and only if a character of the group generated by the unit vectors maps them into $[1/3, 2/3]$. For $F = \mathbb{Q}(\sqrt{d})$ and finitely many unit vectors the last condition is a question on a 4-dimensional torus, which exact branch-and-bound certificates settle. We obtain $\chi(\mathbb{Q}(\sqrt{d})^2) = 4$ for $d = 83, 107, 203$, $4 \leq \chi(\mathbb{Q}(\sqrt{143})^2) \leq 5$ and $\chi(\mathbb{Q}(\sqrt{167})^2) \geq 4$: five fields for which our searches for finite non-3-colourable unit-distance graphs had failed. The theorem for finite groups, the answers to the questions of Glasscock, Koutsogiannis and Richter and of Liu, and the lonely runner theorem for three speeds that the latter uses are formally verified in Lean.

1. INTRODUCTION

For an abelian group Γ and a nonempty subset $S = -S$ of $\Gamma \setminus \{0\}$, the Cayley graph $\text{Cay}(\Gamma, S)$ has vertex set Γ and an edge between g and $g+s$ for $g \in \Gamma$, $s \in S$. A *character* is a homomorphism $\xi : \Gamma \rightarrow \mathbb{R}/\mathbb{Z}$. For $x \in \mathbb{R}/\mathbb{Z}$ let $\|x\|$ be its distance to 0.

For integers $p \geq 2q \geq 2$ the *circular clique* $K_{p/q}$ has vertex set $\mathbb{Z}/p$, with $i \sim j$ when $q \leq (j - i \bmod p) \leq p - q$. Thus $K_{3/1} = K_3$, $K_{(2k+1)/k} = C_{2k+1}$, and $K_{p/q} \rightarrow K_{p'/q'}$ if and only if $p/q \leq p'/q'$; the circular chromatic number $\chi_c(G)$ is the infimum of the p/q with $G \rightarrow K_{p/q}$, and $\chi(G) - 1 < \chi_c(G) \leq \chi(G)$ [43].

Theorem 1. *Let $p \geq 2q \geq 2$ with $p < 4q$. The graph $\text{Cay}(\Gamma, S)$ has a homomorphism to $K_{p/q}$ if and only if there is a character ξ of Γ with $\xi(s) \in [q/p, 1 - q/p]$ for every $s \in S$. In particular, $\text{Cay}(\Gamma, S)$ is 3-colourable if and only if some character maps every element of S into $[1/3, 2/3]$, and it maps to C_{2k+1} if and only if some character maps every element of S into $[k/(2k+1), (k+1)/(2k+1)]$.*

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Let $\kappa(S) = \sup_{\xi} \inf_{s \in S} \|\xi(s)\|$; the supremum is attained, as the dual group is compact and $\xi \mapsto \inf_s \|\xi(s)\|$ is upper semicontinuous. The *if* direction of Theorem 1 is the classical inequality $\chi_c \leq 1/\kappa$: for distance graphs it is Zhu's multiplier method, surveyed by Liu [32, §2], and for Cayley graphs of $\mathbb{Z}/p$ it appears in the proof of [3, Theorem 3.2] and the remark after it. The content of the theorem is the converse:

Corollary 2. *If $\chi_c(\text{Cay}(\Gamma, S)) < 4$, then $\chi_c(\text{Cay}(\Gamma, S)) = 1/\kappa(S)$.*

Proof. For every rational p/q with $\chi_c < p/q < 4$ the graph maps to $K_{p/q}$, so by Theorem 1 $\kappa(S) \geq q/p$; hence $1/\kappa \leq \chi_c$, and $\chi_c \leq 1/\kappa$ always. $\square$

The bound 4 cannot be raised: $K_4 = \text{Cay}((\mathbb{Z}/2)^2, (\mathbb{Z}/2)^2 \setminus \{0\})$ has $\chi_c = 4$, but every character of $(\mathbb{Z}/2)^2$ vanishes on a nonzero element, so $\kappa = 0$ (Section 6).

The proof uses the winding of circular colourings. A homomorphism to $K_{p/q}$ lifts each edge to an integer in $[q, p - q]$, and the sum of the lifts along a closed walk is a multiple of p (Brewster, McGuinness, Moore and Noel [8, §2.1]); for $p < 4q$ the two sides of a square cannot differ by a whole turn, so this wind does not change under homotopy (Brewster and Moore [9, §2]). For K_3 it is the discrete winding number of Krebs and Sankar [31] (their Definition 3.5, Propositions 3.3 and 3.6, Remark 3.7), which they use to show that groups of exponent 4 have no 3-chromatic Cayley graphs. What is new, as far as we know, is the averaging of the winding over the group with an invariant mean, and the characterisation that results (Section 7).

For K_3 the theorem has a reading in terms of recurrence. A set $S \subseteq \Gamma$ is a set of *r-chromatic recurrence* if every r -colouring of Γ has two points of one colour that differ by an element of S , and a set of *Bohr recurrence* if it meets every Bohr neighbourhood of 0, that is, every set $\{g : \|\xi_1(g)\| < \varepsilon, \dots, \|\xi_m(g)\| < \varepsilon\}$ with characters ξ_j and $\varepsilon > 0$ [25, Definitions 1.1 and 1.2]. Katznelson's question [29, 22, 25] asks whether every set of Bohr recurrence is a set of r -chromatic recurrence for every r . It is open, and the answer is not known for any countably infinite abelian group [25]. For $r = 2$ the answer is easily yes, and Alweiss notes that a counterexample needs at least three colours [2, §3]. Theorem 1 gives $r = 3$:

Corollary 3. *A set $S \subseteq \Gamma \setminus \{0\}$ is a set of 3-chromatic recurrence if and only if it meets the set $\{g : \|\xi(g)\| < 1/3\}$ for every character ξ . In particular, every set of Bohr recurrence is a set of 3-chromatic recurrence, and even $\chi_c(\text{Cay}(\Gamma, S \cup -S)) \geq 4$ for such a set; every 3-colourable abelian Cayley graph has a 3-colouring whose colour classes are $\xi^{-1}[0, 1/3)$, $\xi^{-1}[1/3, 2/3)$ and $\xi^{-1}[2/3, 1)$ for a single character ξ . For every 3-colouring $\mathbb{N} = A_1 \cup A_2 \cup A_3$ there is $\alpha \in \mathbb{R}/\mathbb{Z}$ with*

$$(A_1 - A_1) \cup (A_2 - A_2) \cup (A_3 - A_3) \supseteq \{n \in \mathbb{N} : \|n\alpha\| < 1/3\};$$

in particular this union is a Bohr set.

Proof. Apply Theorem 1 with $p/q = 3$ to $S \cup (-S)$; the colouring is the one in the *if* direction of the proof. A set of Bohr recurrence has $\kappa = 0$, so Corollary 2 gives $\chi_c \geq 4$. For the last statement let $S = \mathbb{Z} \setminus \bigcup_i (A_i - A_i)$; then $S = -S$, since each $A_i - A_i$ is symmetric, and $0 \notin S$, since some A_i is nonempty. Every finite subgraph of $\text{Cay}(\mathbb{Z}, S)$ has a translate inside $\mathbb{N}$, where the given colouring is proper, so $\text{Cay}(\mathbb{Z}, S)$ is 3-colourable [16], and Theorem 1 gives α with $\|s\alpha\| \geq 1/3$ for all $s \in S$. $\square$

The radius $1/3$ is best possible. For irrational α , the colouring $n \mapsto [3\{n\alpha\}]$ of $\mathbb{N}$ has $\bigcup_i (A_i - A_i) \cap \mathbb{N} = U := \{n \geq 1 : \|n\alpha\| < 1/3\}$, and U contains no set $\{n \geq 1 : \|n\beta\| < \delta\}$ with $\delta > 1/3$. Indeed, let H be the closure of the multiples of (α, β) in $(\mathbb{R}/\mathbb{Z})^2$; the multiples with $n \geq 1$ are dense in H , so it suffices to find $(x, y) \in H$ with $\|x\| > 1/3$ and $\|y\| < \delta$. If $H = (\mathbb{R}/\mathbb{Z})^2$, take $(1/2, 0)$. Otherwise $H = \{(x, y) : ax + by = 0\}$ for a generator (a, b) of the annihilator of H in $\mathbb{Z}^2$, and $b \neq 0$ because α is irrational. If $|b| \geq 2$, some $(1/2, y) \in H$ has $\|y\| \leq 1/4$. If $b = \pm 1$, then $H = \{(x, cx)\}$ for an integer c , and $x = 1/3 + \varepsilon$ with small $\varepsilon > 0$ has $\|cx\| \leq 1/3 + |c|\varepsilon < \delta$.

For $\Gamma = \mathbb{Z}$ the invariant mean can be avoided. If S is finite with largest element L , two windows $[i, i + L)$ and $[j, j + L)$, $i < j$, of a 3-colouring of $\text{Cay}(\mathbb{Z}, S)$ carry the same colours, and repeating the colouring on $[i, j)$ with period $N = j - i$ gives a 3-colouring of $\text{Cay}(\mathbb{Z}/N, S)$;

Theorem 1 for the finite group $\mathbb{Z}/N$, where the mean is a plain average, gives $\alpha \in \frac{1}{N}\mathbb{Z}/\mathbb{Z}$. For infinite S , a limit point of the characters obtained for finite $S_m \uparrow S$ works, since the conditions $\|s\alpha\| \geq 1/3$ are closed. So the last statement of Corollary 3 needs only the finite cyclic case of Theorem 1; this is the proof that is checked in Lean (Remark 6).

The last statement answers Question 3 of Glasscock, Koutsogiannis and Richter [22], who proved it for two colours [22, Theorem 4.9] and wrote that the case of three colours was not known. As they remark, it follows that the difference set of a set $A \subseteq \mathbb{N}$ with $A \cup (A - \ell_1) \cup (A - \ell_2) \supseteq \mathbb{N}$ is a Bohr₀ set. In the same way, if three translates of $A \subseteq \mathbb{Z}^d$ cover $\mathbb{Z}^d$, then $A - A$ contains $\{z : \|z \cdot \lambda\| < 1/3\}$ for some $\lambda \in (\mathbb{R}/\mathbb{Z})^d$, the case of three translates of [22, Question 8]. By Corollary 3, a counterexample to Katznelson's question needs at least four colours, and with four colours it would be a Cayley graph with $\chi_c = \chi = 4$. We have not found the case $r = 3$ in [22, 25, 27, 34, 2, 7], nor in a second search of the literature; we have not been able to read [29]. Glasscock, Koutsogiannis and Richter, who cite [29], state that the case of three colours is open [22].

A second consequence concerns periodicity. A colouring of a Cayley graph is *periodic* if it is invariant under a subgroup of finite index.

Corollary 4. *Let Γ be a finitely generated abelian group and S finite. If $\text{Cay}(\Gamma, S)$ has a homomorphism to $K_{p/q}$ with $p/q < 4$ (for instance, if it is 3-colourable), then it has a periodic one, and whether it has one is decidable. Whether $\chi_c(\text{Cay}(\Gamma, S)) < 4$ is decidable, and if it is, $\chi_c = 1/\kappa(S)$ is a rational number that can be computed, and there is a periodic homomorphism to K_{χ_c} .*

Proof. Write $\Gamma \cong \mathbb{Z}^r \times F$ with F finite. In these coordinates the characters with $\xi(S) \subseteq [q/p, 1 - q/p]$ form a finite union of polytopes with rational vertices in the torus $(\mathbb{R}/\mathbb{Z})^r \times \widehat{F}$ (one for each character of F and each choice of the integer parts of the values $\xi(s)$). If one exists, a rational one exists; it has finite order, and the colouring of the *if* direction of the proof is constant on the cosets of its kernel, a subgroup of finite index. The vertices can be enumerated. In the same way $\kappa(S)$ is the maximum of a piecewise linear function with rational data over finitely many rational polytopes, so it is rational and computable, and it is attained at a character of finite order; and $\chi_c < 4$ if and only if $\kappa(S) > 1/4$, by Corollary 2. In that case the colouring of the *if* direction for such a character is a periodic homomorphism to $K_{1/\kappa(S)} = K_{\chi_c}$. $\square$

For $\Gamma = \mathbb{Z}$ and finite S (distance graphs) there are periodic optimal colourings for any number of colours, by a pigeonhole argument. For $\mathbb{Z}^d$ with $d \geq 2$ this argument fails, and subshifts of finite type on $\mathbb{Z}^2$ can be aperiodic. Abrishami, Esperet, Giocanti, Hamann, Knappe and Möller ask whether every locally finite graph with a periodic proper colouring has a periodic proper colouring with $\chi(G)$ colours [1, Problem 4.6]; they answer yes for graphs of bounded pathwidth, such as Cayley graphs of 2-ended groups, and no for some ∞ -ended graphs. Corollary 4 answers yes for Cayley graphs of finitely generated abelian groups with finite connection sets and $\chi \leq 3$ (Theorem 1 with $p/q = 2$ covers $\chi = 2$). Vallentin, Weißbach and Zimmermann note that it is not known whether the chromatic number of a lattice, that is, of the Cayley graph of $\Lambda \cong \mathbb{Z}^n$ on its strict Voronoi vectors, is computable, and ask whether a periodic colouring with $\chi(\Lambda)$ colours always exists [42]. By Corollary 4, whether $\chi(\Lambda) \leq 3$ is decidable (given the strict Voronoi vectors in coordinates of Λ), and if it is, a periodic 3-colouring exists. Three is special: by the theorem of Greenfeld and Tao on the periodic tiling conjecture [23], some Cayley graphs of $\mathbb{Z}^d$ have no periodic colouring with χ colours (Proposition 14).

Section 2 proves the theorem. Section 3 derives Payan's theorem [38] and the exponent-4 case of Krebs and Sankar [31]. Section 4 treats distance graphs. Section 5 applies the theorem to the unit-distance graph of the plane over a number field, the setting that led to it, and Section 6 records that nothing of the kind holds for four colours, and that periodicity fails for many colours.

2. PROOF OF THEOREM 1

If. Let $c(g) = \lfloor p\xi(g) \rfloor \bmod p$, where $\xi(g)$ is read in $[0, 1)$. Along an edge, $p\xi$ moves by an amount $a \in [q, p-q]$ modulo p , and $\lfloor y+a \rfloor - \lfloor y \rfloor \in \{\lfloor a \rfloor, \lceil a \rceil\} \subseteq [q, p-q]$. So c is a homomorphism to $K_{p/q}$.

Only if. Let $c : \Gamma \rightarrow \mathbb{Z}/p$ be a homomorphism to $K_{p/q}$, and let Γ' be the subgroup generated by S . A character of Γ' extends to Γ , since $\mathbb{R}/\mathbb{Z}$ is divisible, so it suffices to find the character on Γ' . We work in Γ' because $\text{Cay}(\Gamma', S)$ is connected, which Step 2 uses: averaging over all of Γ would fail when $\text{Cay}(\Gamma, S)$ is disconnected, since different components can wind in opposite directions. For $g \in \Gamma'$ and $s \in S$ let $\delta(g, s) \in \{q, \dots, p-q\}$ be the representative of $c(g+s) - c(g)$. Then $\delta(g+s, -s) = p - \delta(g, s)$.

Step 1: squares do not wind. For $s, t \in S$ the two integers $\delta(g, s) + \delta(g+s, t)$ and $\delta(g, t) + \delta(g+t, s)$ lie in $[2q, 2p-2q]$ and are congruent to $c(g+s+t) - c(g)$ modulo p . They differ by at most $2p-4q$, which is less than p because $p < 4q$; so they are equal. For K_3 this is [31, Proposition 3.6]; for $2 < p/q < 4$ it is the fact that a 4-cycle has wind 2 [9, §2].

Step 2: the winding of a closed walk (compare [31, Definition 3.5, Propositions 3.3 and 3.6] for K_3 and [9, §2] for $K_{p/q}$). For a walk from g with steps $s_1, \dots, s_N$ let W be the sum of δ along it, $W = \sum_j \delta(g + s_1 + \dots + s_{j-1}, s_j)$. By Step 1, W does not change when two consecutive steps are exchanged, so it depends only on g and on the steps up to order. If the walk is closed ($\sum s_j = 0$), then $W \equiv 0 \pmod{p}$, and W does not depend on g : the closed walk from g with steps $s, s_1, \dots, s_N, -s$ has sum $\delta(g, s) + W(g+s) + \delta(g+s, -s) = W(g+s) + p$, and, moving its first step to the end, also $W(g) + p$; and $\text{Cay}(\Gamma', S)$ is connected.

Step 3: averaging. As Γ' is abelian, it is amenable: there is a mean m on the bounded functions on Γ' that is invariant under translations. For finite Γ' it is the plain average; for countable Γ' one can take a Banach limit of the averages over a Følner sequence. (Alternatively, one can prove the theorem for finitely generated subgroups of Γ' and pass to Γ' by compactness of the dual group.) Let

$$f(s) = m(g \mapsto \delta(g, s)) \in [q, p-q].$$

By invariance, $f(-s) = m(g \mapsto p - \delta(g-s, s)) = p - f(s)$. For a closed walk $(s_1, \dots, s_N)$, the function $g \mapsto W(g) = \sum_j \delta(g + s_1 + \dots + s_{j-1}, s_j)$ is a constant W by Step 2; applying m , which is linear and invariant under translations, gives

$$\sum_{j=1}^N f(s_j) = W \in p\mathbb{Z}.$$

Step 4: the character. Put $\xi(s) = f(s)/p \in [q/p, 1 - q/p]$; then $\xi(-s) = 1 - \xi(s)$, and if $2s = 0$ then $\xi(s) = 1/2$. Let Z be the abelian group generated by symbols $[s]$, $s \in S$, subject to $[-s] = -[s]$, and R the kernel of $Z \rightarrow \Gamma'$, $[s] \mapsto s$. Then $[s] \mapsto \xi(s)$ is well defined modulo 1 on Z , and an element of R is the class of a closed walk, on which $\sum_j \xi(s_j) = W/p \in \mathbb{Z}$. So ξ vanishes on R modulo 1 and defines a character of $Z/R = \Gamma'$ with $\xi(S) \subseteq [q/p, 1 - q/p]$. $\square$

Only Step 1 uses $p < 4q$. For $p = 3$, $q = 1$ the lift $\delta \in \{1, 2\}$ is the sign of Krebs and Sankar shifted by $3/2$: $\sigma = 3 - 2\delta = \pm 1$.

Remark 5. For $\Gamma = \mathbb{Z}^2$ with the standard generators this is the height-function picture of 3-colourings of the square lattice, in which the height changes by ± 1 along edges and by 0 around squares (compare [40]); f is the averaged slope. Theorem 1 says that the averaged slopes of any colouring of any abelian Cayley graph form a character as above. The colouring c itself need not come from a character: for $\Gamma = \mathbb{Z}^2$ most 3-colourings do not.

Remark 6 (Formal proof). For finite Γ the theorem is formalised in Lean 4 with Mathlib in `lean/TheoremWplus.lean` of [18]: the *only if* direction for every $p < 4q$, with the character first on the subgroup generated by S and then extended to Γ , and the *if* direction. The *only if* direction for K_3 , with the signs of Krebs and Sankar, is also in `lean/TheoremW.lean`. The formal proofs follow Steps 1–4, with the plain average as the mean. The last statement of Corollary 3 (the answer to Question 3 of [22]) and, for $\Gamma = \mathbb{Z}$, the non-trivial direction of its first

statement are formalised from it in `lean/Recurrence.lean`, by the route through finite cyclic groups described after the corollary. The first two assertions of Corollary 9 are formalised in `lean/DistLiu.lean`: Theorem 1 for $G(\mathbb{Z}, D)$ with D finite (from the finite case, through one periodic window), its converse, Lemma 10, and for $|D| = 3$ the statement that $G(\mathbb{Z}, D)$ maps to $K_{p/q}$ if and only if $\kappa(D) \geq q/p$. These files depend only on Lean’s three standard axioms; the Lean workflow of [18] builds them, compares their axioms with a stored list and replays them in Lean’s kernel.

3. GROUPS OF SMALL EXPONENT

Corollary 7. *Let Γ be an abelian group of exponent 2 or 4. Then no Cayley graph of Γ has circular chromatic number strictly between 2 and 4; in particular none has chromatic number 3.*

Proof. A character of Γ takes values in $\frac{1}{4}\mathbb{Z}/\mathbb{Z}$, and the only such value in $[q/p, 1 - q/p]$, for $p/q < 4$, is $1/2$. So if $\text{Cay}(\Gamma, S)$ maps to $K_{p/q}$ with $p/q < 4$, a character maps S to $1/2$, and $g \mapsto \lfloor 2\xi(g) \rfloor$ is a proper 2-colouring. $\square$

Corollary 7 is a uniform proof of known facts. For exponent 2 and chromatic number 3 it is Payan’s theorem on cube-like graphs [38], which Cervantes and Krebs reproved [12]. The full statement for exponent 2 also follows from known results: Payan’s proof shows that a non-bipartite cube-like graph contains a generalised Mycielski graph of an odd cycle ([38]; see [4, Theorem 1.4]), and these graphs are strongly topologically 4-chromatic (Stiebitz; see [41, Proposition 9]), hence have $\chi_c = 4$ [41, Theorem 6]. For exponent 4 and chromatic number 3 it is a case of the theorem of Krebs and Sankar [31, Theorem 1.2]: an abelian group has a 3-chromatic Cayley graph if and only if its exponent is not 1, 2 or 4. Their proof uses the winding number of 3-colourings: for exponent 4 they show that the discrete fundamental group is torsion, and by their Theorem 4.1 a non-bipartite graph with a torsion fundamental group is not 3-colourable; run with the wind of [9], the same argument gives $\chi_c \geq 4$. The same argument as for Corollary 7 gives, for instance, that a Cayley graph of an abelian group of exponent 8 that maps to C_5 is bipartite, since $1/2$ is the only element of $\frac{1}{8}\mathbb{Z}/\mathbb{Z}$ in $[2/5, 3/5]$. For C_3 the values $3/8$ and $5/8$ are also available, and indeed such groups have 3-chromatic Cayley graphs [31].

4. DISTANCE GRAPHS

For a finite set D of positive integers, the distance graph $G(\mathbb{Z}, D)$ is $\text{Cay}(\mathbb{Z}, \pm D)$. Its characters are $x \mapsto \alpha x$, $\alpha \in \mathbb{R}/\mathbb{Z}$, and $\kappa(D) = \max_{\alpha} \min_{d \in D} \|\alpha d\|$ is the parameter of the lonely runner problem [32, 39].

Corollary 8. *$G(\mathbb{Z}, D)$ is 3-colourable if and only if $\kappa(D) \geq 1/3$; it maps to $K_{p/q}$, $p/q < 4$, if and only if $\kappa(D) \geq q/p$; and if $\chi_c(G(\mathbb{Z}, D)) < 4$, then $\chi_c(G(\mathbb{Z}, D)) = 1/\kappa(D)$.*

The set of α with $\|\alpha d\| \geq 1/3$ for all $d \in D$ is a finite union of closed intervals whose endpoints are among the points $(j + 1/3)/d$ and $(j + 2/3)/d$. So it is non-empty if and only if it contains one of these points, and the test is finite. We compared it, for all 1 747 sets $D \subseteq [1, 24]$ with $|D| = 3$ and $\gcd(D) = 1$, with 3-colourability of the segment $[0, 12 \max D + 59]$ decided by a SAT solver. They agree, and the 74 sets that need four colours are exactly those of Zhu’s classification [44] of 3-element sets with $\chi(G(\mathbb{Z}, D)) = 4$, namely $\{1, 2, 3m\}$ and $\{a, b, a + b\}$ with $a \not\equiv b \pmod{3}$ (after division by the gcd). Liu’s survey asks whether $\chi_c(G(\mathbb{Z}, D)) < 1/\kappa(D)$ can happen when $|D| = 3$ [32, Problem 3]. It cannot:

Corollary 9. *For every set D of three positive integers, $\chi_c(G(\mathbb{Z}, D)) = 1/\kappa(D)$ and $\chi(G(\mathbb{Z}, D)) = \lceil 1/\kappa(D) \rceil$. In particular $\chi_c(G(\mathbb{Z}, D)) \leq 4$, with equality if and only if $D = \{k, 2k, 3k\}$ for some k .*

Proof. If $\chi_c < 4$, the first equality is Corollary 8. Otherwise $4 \leq \chi_c \leq 1/\kappa(D) \leq 4$, the last inequality being the lonely runner theorem for three speeds [5, 14] (Lemma 10 below). The colouring in the proof of the *if* direction of Theorem 1, which is valid for every $p/q \geq 2$, gives a homomorphism to $K_{1/\kappa(D)}$ ($\kappa(D)$ is rational), so $\chi \leq \lceil 1/\kappa(D) \rceil$, and $\chi \geq \chi_c$. Finally, $\kappa(D) = 1/4$

only for $D = \{k, 2k, 3k\}$, the only tight instances of the lonely runner problem with three speeds [15], [39, §§4 and 6.2]. $\square$

So the chromatic number of a distance graph with three distances is $\lceil 1/\kappa(D) \rceil$, a formula in terms of the lonely runner parameter. As a check, for all 28 876 sets $D \subseteq [1, 60]$ with $|D| = 3$ and $\gcd(D) = 1$, $\lceil 1/\kappa(D) \rceil$, with $\kappa(D)$ computed exactly, agrees with Zhu's formula for $\chi(G(\mathbb{Z}, D))$ [44], and $\kappa(D) = 1/4$ only for $D = \{1, 2, 3\}$.

The lonely runner theorem for three speeds is classical [5, 14]; we include the short proof that is formalised (Remark 6).

Lemma 10. *For all real $a, b, c > 0$ there is $t \in \mathbb{R}$ with $\|at\| \geq 1/4$, $\|bt\| \geq 1/4$ and $\|ct\| \geq 1/4$.*

Proof. We may assume $0 < a \leq b \leq c$. Put $P = c/a$, $Q = c/b$ and $\rho = P/Q = b/a$, so $P \geq Q \geq 1$ and $\rho \geq 1$. For an odd integer x let $t = x/(4c)$. Then $\|ct\| = \|x/4\| = 1/4$; $\|at\| \geq 1/4$ if $P \leq x \leq 3P$, since then $at = x/(4P) \in [1/4, 3/4]$; and $\|bt\| \geq 1/4$ if $(4m+1)Q \leq x \leq (4m+3)Q$ for some integer $m \geq 0$, since then $bt \in [m + 1/4, m + 3/4]$. Every closed interval of length at least 2 contains an odd integer.

- If $\rho \geq 3$, let $m = \lceil (\rho - 1)/4 \rceil$. Then $P = \rho Q \leq (4m+1)Q$ and $(4m+3)Q < (\rho+6)Q \leq 3\rho Q = 3P$, so $[(4m+1)Q, (4m+3)Q]$, of length $2Q \geq 2$, lies in $[P, 3P]$.
- If $7/3 \leq \rho < 3$, the same holds with $m = 1$: $P < 5Q$ and $7Q \leq 3P$.
- If $\rho < 7/3$, an odd integer in $[P, 3Q]$ (with $m = 0$) or in $[5Q, 3P]$ (with $m = 1$) works. If both intervals are shorter than 2, then $4Q = 3(3Q - P) + (3P - 5Q) < 8$, so $Q < 2$ and $P < (2+5Q)/3 < 4$. Then $x = 3$ works if $P \leq 3$; $x = 5$, with $m = 0$, if $P > 3$ and $3Q \geq 5$; and $x = 9$, with $m = 1$, if $P > 3$ and $3Q < 5$, since then $9/7 < 3P/7 < Q < 5/3$. $\square$

For larger sets $\chi_c(G(\mathbb{Z}, D)) < 1/\kappa(D)$ does happen [32, Theorems 17 and 18], but by Corollary 8 only when $\chi_c \geq 4$; indeed, in that family the sets with $\chi_c < 4$ satisfy $\chi_c = 1/\kappa$, as one checks from the formulas there. The equality $\chi_c = 1/\kappa$ also reduces the computation of $\chi_c(G(\mathbb{Z}, D))$, when it is less than 4, to that of $\kappa(D)$, whose denominator divides a sum of two elements of D [32].

5. PLANES OVER NUMBER FIELDS

Let F be a field of characteristic 0 with $i \notin F$, and $L = F(i)$. Identify F^2 with L by $(x, y) \mapsto x + iy$; the unit vectors are $T = \{z \in L : z\bar{z} = 1\}$, and the unit-distance graph of F^2 is $\text{Cay}(L, T)$, a disjoint union of copies of $\text{Cay}(A, T)$ with $A = \mathbb{Z}[T]$.

Corollary 11. *$\chi(F^2) \leq 3$ if and only if some character of A maps every unit vector into $[1/3, 2/3]$. If for a finite set $U \subset T$ no character of the group $\mathbb{Z}U$ maps U into $[1/3, 2/3]$, then $\chi(F^2) \geq 4$, and some finite unit-distance graph over F is not 3-colourable.*

The last assertion is the theorem of de Bruijn and Erdős [16]. If F is a number field, then $\mathbb{Z}U$ is a finitely generated subgroup of the $\mathbb{Q}$ -vector space L , hence free of rank at most $[L : \mathbb{Q}]$; for a quadratic field F the test for a finite U is a question on a torus of dimension at most 4. A locally constant 3-colouring at a place of F [17, 36, 19] is a colouring of a finite quotient of A , and Theorem 1 applied to that quotient gives such a character.

5.1. Real quadratic fields. For $F = \mathbb{Q}(\sqrt{d})$ with $d > 1$ squarefree, $\chi(F^2) = 2$ if $d \not\equiv 3 \pmod{4}$ [28, 17, 36]. If $d \equiv 3 \pmod{4}$, then $\chi(F^2) \leq 3$ when $d \not\equiv 2 \pmod{3}$ [17, 36], and $\chi(F^2) \leq 4$ when $d \equiv 3 \pmod{8}$ [17, Theorem 10] or $d \equiv 0, 1, 2, 4 \pmod{7}$ [36]. In [20] we found finite non-3-colourable unit-distance graphs over $\mathbb{Q}(\sqrt{d})$ for 27 values $d \equiv 11 \pmod{12}$ below 1000, but our searches for such graphs failed for $d = 83, 107, 143, 167, 203$, the values that the local-global question of [19] predicted to need four colours and that remained open below 210.

Fix a denominator D and let U_D be the unit vectors $((a+b\sqrt{d})/D, (c+e\sqrt{d})/D)$, $a, b, c, e \in \mathbb{Z}$, one of each pair $\pm u$. A character with $\xi(U_D) \subseteq [1/3, 2/3]$ gives $f \in [1/3, 2/3]^{U_D}$ (f_u the lift of $\xi(u)$ in $[1/3, 2/3]$) with $\langle r, f \rangle \in \mathbb{Z}$ for every integer relation r among the vectors of U_D . Using an LLL-reduced basis of the relations, we rule this out by branch and bound over the integers $\langle r, f \rangle$: each leaf carries a rational vector y (a Farkas certificate) such that $\langle y, \cdot \rangle$ applied to the equations

fixed at that leaf cannot hold on the box $[1/3, 2/3]^{U_D}$. A short program checks a certificate in exact integer and rational arithmetic: the vectors have length 1 ($a^2 + db^2 + c^2 + de^2 = D^2$, $ab + ce = 0$), the relations hold, every branch covers every integer value in its range, and every Farkas vector is valid. Any set of valid relations gives necessary conditions, so infeasibility is a proof whether or not the relations form a basis.

Theorem 12. $\chi(\mathbb{Q}(\sqrt{d})^2) = 4$ for $d = 83, 107, 203$; $4 \leq \chi(\mathbb{Q}(\sqrt{143})^2) \leq 5$; and $\chi(\mathbb{Q}(\sqrt{167})^2) \geq 4$.

Proof. The lower bounds are Corollary 11 with the certificates of Table 1 [18]. The upper bound 4 holds because $83 \equiv 107 \equiv 203 \equiv 3 \pmod{8}$ [17, Theorem 10]; this reduction at 2 is also formalised in `lean/QuadraticPlanes.lean` of [18]. For $d = 143$ the place above 11 is ramified in $\mathbb{Q}(\sqrt{143})$ with residue field $\mathbb{F}_{11}$, and i is not in its completion; unit vectors are integral there and reduce to the circle $x^2 + y^2 = 1$ of $\mathbb{F}_{11}^2$, whose graph is 5-colourable [20], which colours each coset of the valuation ring. $\square$

d	D	vectors	relations	nodes	leaves
83	510	54	50	6 886	4 664
107	1170	90	86	12 785	9 618
143	1740	54	50	3 861	2 710
167	1560	54	50	8 552	6 094
203	1530	78	74	1 072	732

TABLE 1. Certificates that no character of $\mathbb{Z}U_D$ maps U_D into $[1/3, 2/3]$ (vectors counted up to sign). In each case the vectors span a group of rank 4 and the relations listed generate all relations.

With the values of [20], $\chi(\mathbb{Q}(\sqrt{d})^2)$ is now known for every squarefree $d < 143$ except $d = 47$. $\mathbb{Q}(\sqrt{167})$ is the first admissible field of [19]: no place has a locally constant 4-colouring, so $\chi(\mathbb{Q}(\sqrt{167})^2) \geq 4$ is also the three-colour case of the local–global question there. For $d = 203$ the result shows that no level of the place above 7 is 3-colourable, which [19] had checked only up to level 3. For $d = 83$ and $D = 510$, colouring-guided growth had stopped at 6 008 points because a 3-colouring of the ball of radius 2 extended to all 215 478 candidate points. By greedy deletion, 28 of the 54 vectors already suffice for $d = 83$ (we checked a certificate for this subset but did not keep it). For $\mathbb{Q}(\sqrt{11})$, a subset of 16 vectors with $D = 30$ suffices, in agreement with the 76-vertex graph of [20].

The lower bounds rest on Theorem 1, whose averaging step is not constructive, and on the certificates; we give no explicit finite graph for the five fields. We expect such graphs to be large, which would explain why the searches failed, but this is speculation: Step 3 says nothing about their size. A quantitative version of Step 3 over Følner boxes, or an explicit certified graph, would remove the dependence on the infinite averaging.

All five denominators are even, as they must be. For odd D every vector of U_D has $a + b + c + e$ odd, because $a^2 + db^2 + c^2 + de^2 = D^2$ and d is odd; so $x \mapsto \frac{1}{2}(a + b + c + e)$, for $Dx = (a + b\sqrt{d}) + i(c + e\sqrt{d})$, is a character of $\mathbb{Z}U_D$ with the value 1/2 on U_D , and $\text{Cay}(\mathbb{Z}U_D, U_D)$ is bipartite. The infeasible denominators we found are all multiples of 6, but some multiples of 6 are feasible (for instance $D = 1170$ for $d = 203$). As controls, the test is feasible for 40 pairs (d, D) with $d \in \{2, 3, 5, 6, 7, 15, 19, 31, 39, 43\}$, whose planes are known to be 3-colourable, and for the even denominators $(d, D) = (107, 1450)$, $(143, 1752)$, $(167, 1820)$ and $(203, 1590)$, whose unit vectors also span a group of rank 4.

Question 13. Is $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 4$ for every squarefree $d \equiv 11 \pmod{12}$? In terms of Corollary 11 and [19]: for real quadratic fields, does a character that maps all unit vectors into $[1/3, 2/3]$ exist only when a place gives a locally constant 3-colouring?

6. FOUR AND MORE COLOURS

Step 1 fails for $K_4 = K_{4/1}$, and so does the theorem: $K_4 = \text{Cay}((\mathbb{Z}/2)^2, (\mathbb{Z}/2)^2 \setminus \{0\})$ is 4-colourable, but every character of $(\mathbb{Z}/2)^2$ vanishes on some nonzero element, so none maps the three generators into $[1/4, 3/4]$. So the bound $p < 4q$ in Theorem 1 is sharp. The *if* direction of Theorem 1 holds for every $p/q \geq 2$, so a character with $\xi(S) \subseteq [1/4, 3/4]$ still gives a 4-colouring. For the 108 unit vectors of $\mathbb{Q}(\sqrt{47})$ with denominator 240, which underlie the search for a five-chromatic triangle-free unit-distance graph in [19], a floating-point computation finds none; we have not certified this.

For many colours even Corollary 4 fails, because colourings can encode tilings.

Proposition 14. *There are a finitely generated abelian group Γ (one can take $\Gamma = \mathbb{Z}^d$ for some d , or $\Gamma = \mathbb{Z}^2 \times G_0$ with G_0 finite) and a finite symmetric $S \subseteq \Gamma \setminus \{0\}$ such that $\text{Cay}(\Gamma, S)$ has a proper colouring with $k = \chi(\text{Cay}(\Gamma, S)) < \infty$ colours but no periodic one.*

Proof. For a finite $F \subseteq \Gamma$ let $S_F = (F - F) \setminus \{0\}$. Every translate $x + F$ is a clique of $\text{Cay}(\Gamma, S_F)$ with $|F|$ vertices, so $\chi \geq |F|$, and in a proper colouring with $|F|$ colours every colour class A meets every translate $x + F$ exactly once, that is, $A \oplus (-F) = \Gamma$. Conversely, if $F \oplus A = \Gamma$, giving $f + a$ the colour f is a proper colouring. A periodic colouring gives a periodic tiling (a union of finitely many cosets of a subgroup of finite index), and conversely. Greenfeld and Tao [23, Theorem 1.4 and Corollary 1.6] construct a finite F that tiles $\mathbb{Z}^2 \times G_0$ for some finite abelian G_0 , and one that tiles $\mathbb{Z}^d$ for some d , such that no tiling by F is periodic. Take $S = S_F$ and $k = |F|$. $\square$

For $\Gamma = \mathbb{Z}^d$ one can also make S generate: replace $\mathbb{Z}^d$ by the subgroup generated by $F - F$, which is free abelian, and note that a periodic tiling of it would extend to a periodic tiling of $\mathbb{Z}^d$. Every automorphism of a connected Cayley graph of $\mathbb{Z}^d$ with a finite connection set is affine [35], so a colouring whose colour-preserving automorphisms act with finitely many orbits is invariant under a subgroup of finite index; and $x \mapsto x \bmod N\mathbb{Z}^d$ is a periodic proper colouring once $S \cap N\mathbb{Z}^d = \emptyset$. So the answer to [1, Problem 4.6] is no for Cayley graphs of $\mathbb{Z}^d$, while Corollary 4 gives yes when $\chi \leq 3$. We do not know the least k in Proposition 14; it is at least 4. Nor do we know whether every k -colourable Cayley graph of $\mathbb{Z}^2$ has a periodic k -colouring: a single tile of $\mathbb{Z}^2$ always tiles periodically [6], so the examples above need torsion or $d \geq 3$. Nor do we know whether k -colourability of Cayley graphs of finitely generated abelian groups is decidable for $k \geq 4$: Greenfeld and Tao [24] proved that tiling a periodic subset of $\mathbb{Z}^2 \times G_0$ by one tile is undecidable, but a colouring problem cannot single out a subset. The question of [42] concerns the Voronoi connection set and is not touched.

7. LITERATURE

The closest works are those of Krebs and Sankar [31], whose winding number of 3-colourings is Steps 1 and 2 for K_3 , and of Brewster, McGuinness, Moore and Noel [8] and Brewster and Moore [9], whose wind of (p, q) -colourings is Steps 1 and 2 for $2 < p/q < 4$; they use it for fundamental groups and for reconfiguration, without characters or averaging. Naserasr, Pujol and Wang [37] use winding numbers of circular colourings to prove gaps in χ_c , such as the interval $(2, 4)$ for quadrangulations of the projective plane. The *if* direction of Theorem 1 is classical [32, 3]; see also [21, proof of Theorem 2.3]. The following are consistent with Theorem 1 and do not imply it: Cervantes and Krebs [11, 12, 13] (in [13], 3-colourability for small dimension and rank is characterised by two forbidden subgraphs, diamond lanyards and $C_{13}(1, 5)$, and $C_{13}(1, 5)$ indeed has no suitable character); Krebs and Leyva [30]; the work on Katznelson's question on Bohr recurrence [25, 27, 34, 7], where we found no statement for three colours (we have not seen [29] itself); and Perarnau and Serra [39, §3.5], which has $\chi_f \leq \chi_c \leq 1/\kappa$. We did not find Theorem 1 in these works, in the literature on distance graphs and the lonely runner problem through Liu's survey [32], or in web searches on characters, circular colourings and odd-cycle homomorphisms of Cayley graphs. The proof uses only classical tools, so an earlier occurrence is possible.

The threshold 4 is familiar from reconfiguration. Brewster, McGuinness, Moore and Noel [8] showed that deciding whether two (p, q) -colourings are connected by recolouring one vertex at

a time is polynomial for $2 \leq p/q < 4$ and PSPACE-complete for $p/q \geq 4$, and Brewster and Moore [9] characterised mixing for $p/q < 4$ by the wind of cycles (see also Brewster and Noel [10]); Step 1 is the reason in our setting too. For distance graphs, Haralambis [26] conjectured that $\kappa(D) = \mu(D)$ for $|D| = 3$, where $1/\mu(D)$ is the fractional chromatic number of $G(\mathbb{Z}, D)$ (see [32, Problem 2] and [33, Conjecture 3]), and Liu and Robinson report a computer check for $\max D \leq 25$ [33]; for those sets $\chi_f = \chi_c = 1/\kappa$ follows by sandwiching. Corollary 9 gives $\chi_c = 1/\kappa$ for every D with $|D| = 3$, but says nothing about χ_f .

By Theorem 1, whether a finite abelian Cayley graph, given by Γ and S , maps to $K_{p/q}$ with $p/q < 4$ (for instance, whether it is 3-colourable) can be decided by checking the $|\Gamma|$ characters, with $O(|\Gamma||S|)$ operations: polynomial in the size of the graph.

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